

Scalars, Vectors, Matrices, Tensors:

4 data shapes cover everything in Neural Networks:

Scalar	Vector	Matrix	Tensor
0D •	1D —	2D 	nD 
a single number	a list of numbers	grid of numbers	multi-dim. array
5; 3.14; -50	[5, 30, 15]	$\begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 6 & 7 \end{bmatrix}$	$\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 6 & 7 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 10 & 11 \\ 12 & 13 \end{bmatrix}$
()	(3,)	(3, 2)	(2, 2, 3)
learn.rate; loss val.	word embedding cat → [0.1 0.3 ...]	weight matrix (W) input data (x)	RGB image → (224, 224, 3) (32, 224, 224, 3)